



# **Physically Accurate Boundary Handling in Discrete Cardiac Diffusion**

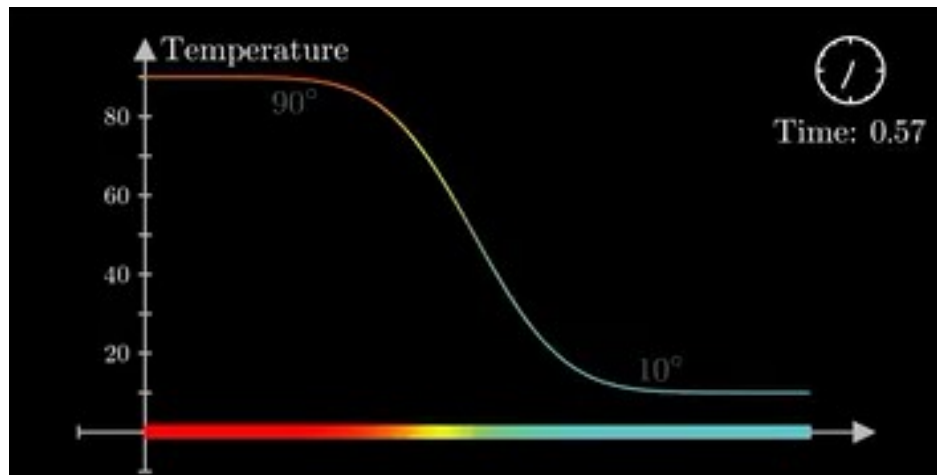
**Charley Chang**



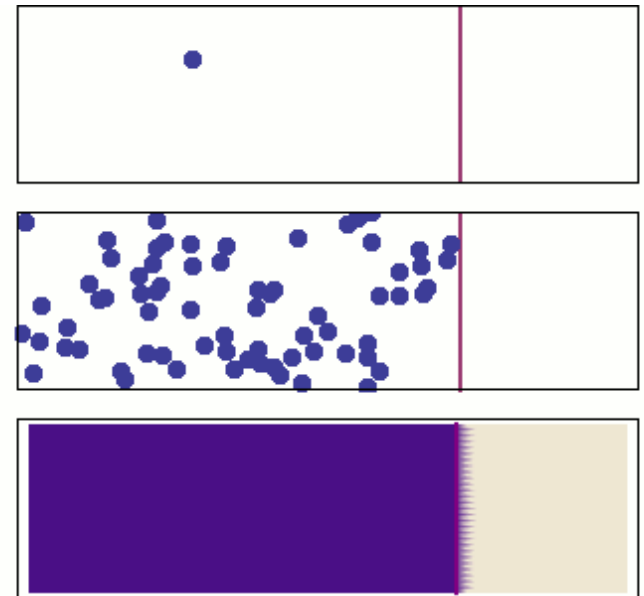
# Fickian Based Diffusion System

Back in 1800s, mathematicians are interested in modeling how diffusive quantities evolves throughout time.

## Modeling of Heat Transfer



## Modeling of Molecular Diffusion



Wikipedia & 3b1b

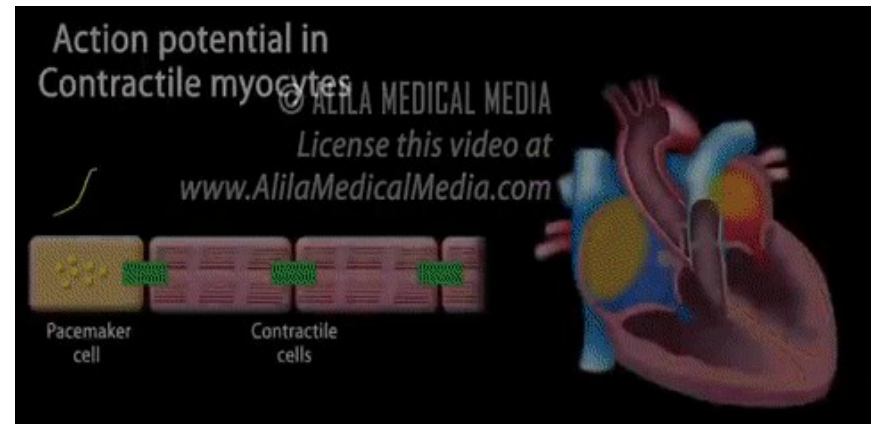
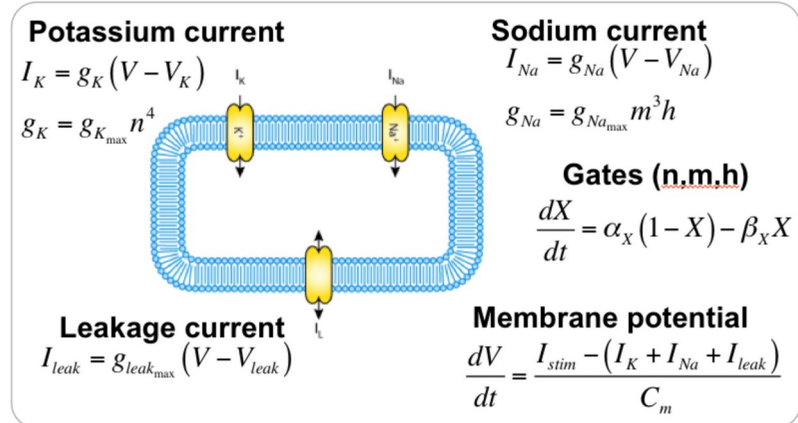
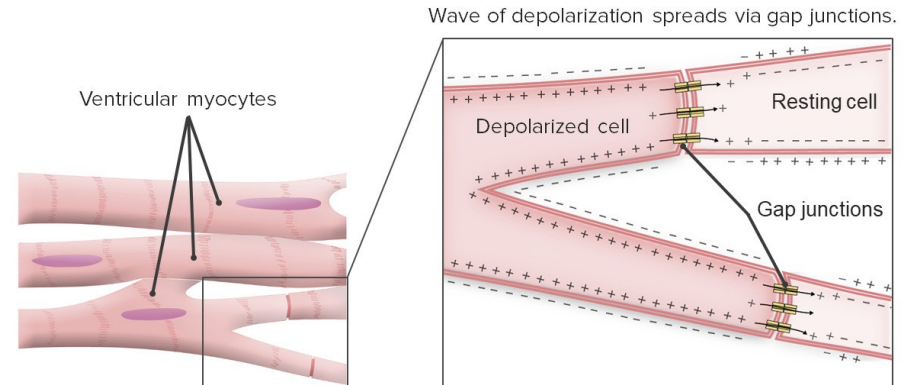
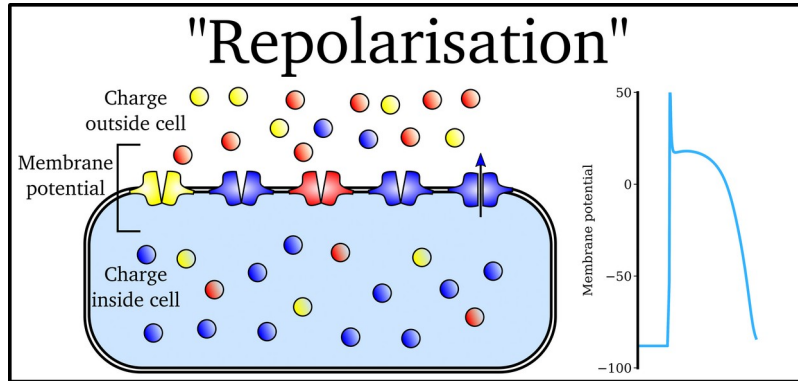
$$j = -D \frac{d\delta}{dx} \quad \frac{d\delta}{dt} = \alpha(\nabla \cdot j)$$

Governing Equations for all Fickian Based System

# Cardiac Conduction Is a Reaction Diffusion System

Cell-Based Ionic Flow Governed By Ionic Channels

Inter-Cell Flow Can be Modeled by Diffusion



AlilaMedicalMedia & Wikipedia

Reaction

Diffusion

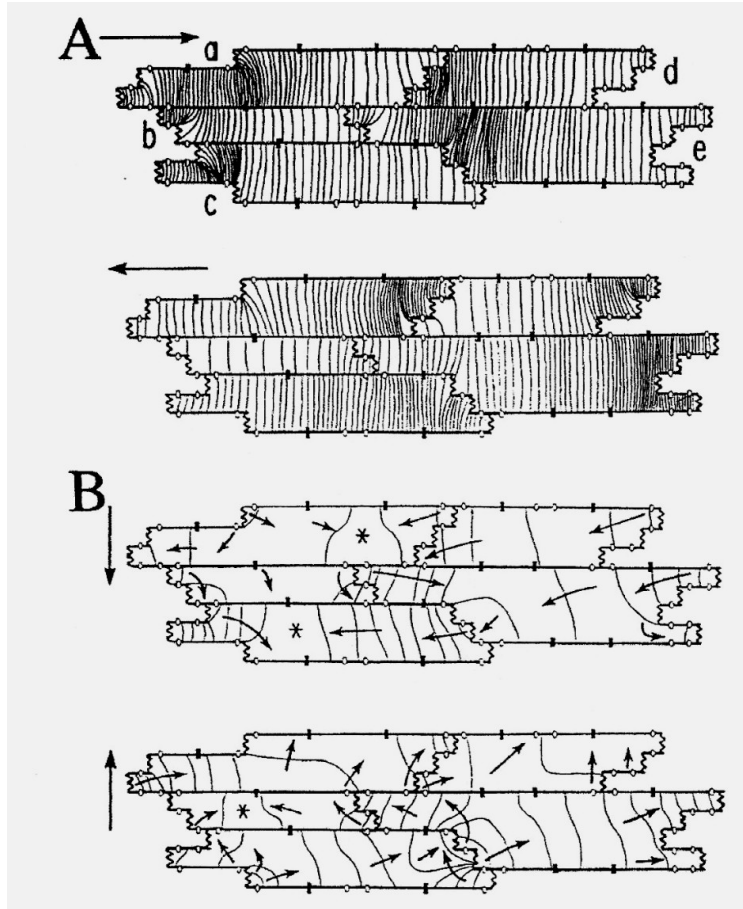
$$\frac{\partial v}{\partial t} - \nabla \cdot (\Sigma_m \nabla v) + I_{ion}(v, \mathbf{w}, \mathbf{z}) = I_{app}(\hat{\mathbf{x}}, t)$$

Cardiac Reaction-Diffusion Equation

# Cardiac Conduction is Also Discrete Diffusion

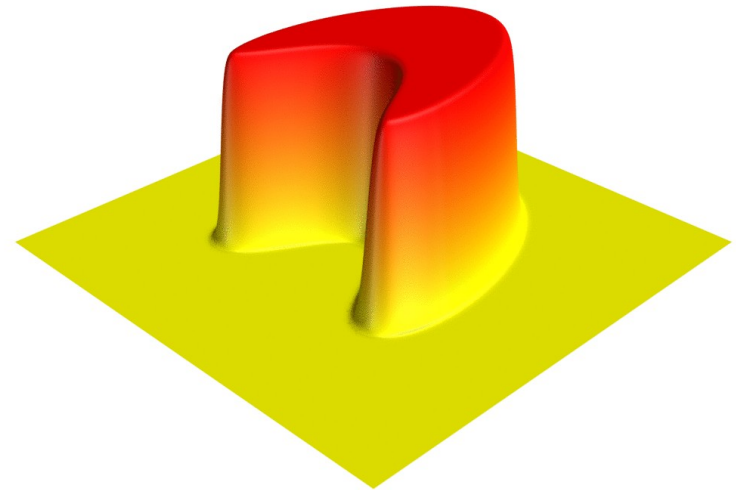
Hi-res cardiac conductions are segmented and non-globally smooth

Isochrone of  $V_m$  During Cardiac Conduction



Rohr et al., 2017

Heat Transfer Simulation of Square Plate



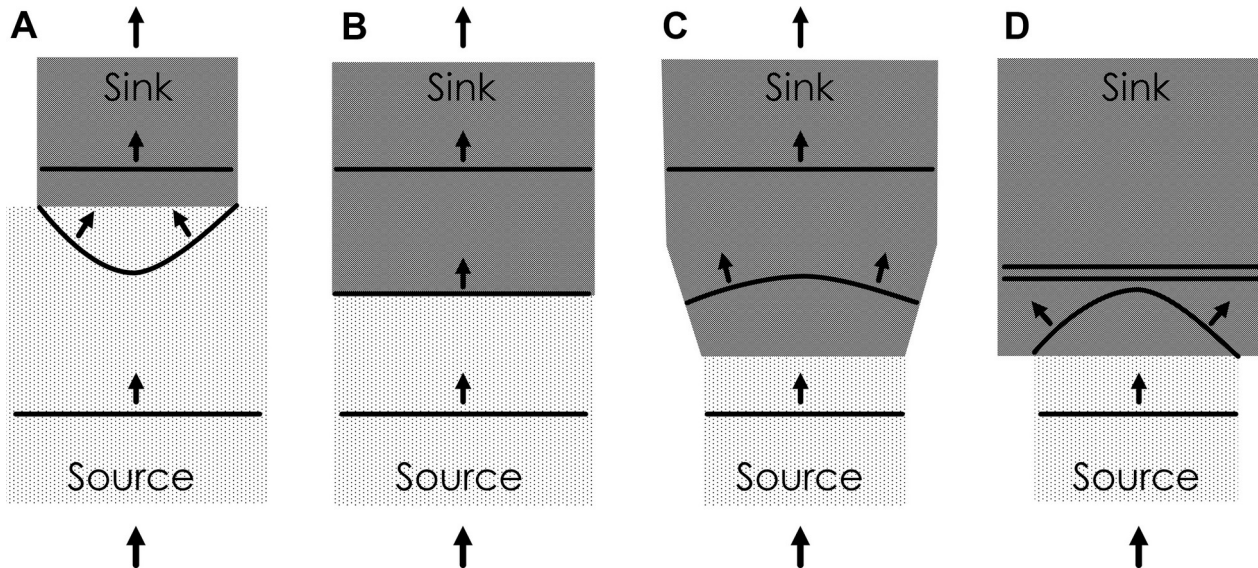
Cardiomyocytes are connected by gap junctions

- Solutions to  $V_m$  inside cells are relatively smooth
- Solutions to diffusion between cells are complex and discrete

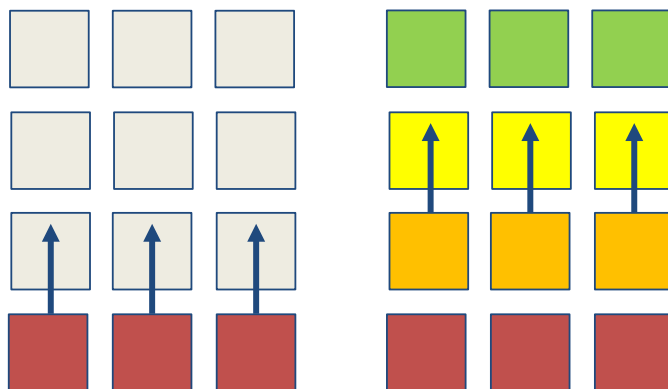
Issue: Heat Equation simulation always smooth out solutions

# Source Sink Mismatch and Wavefront Curvature

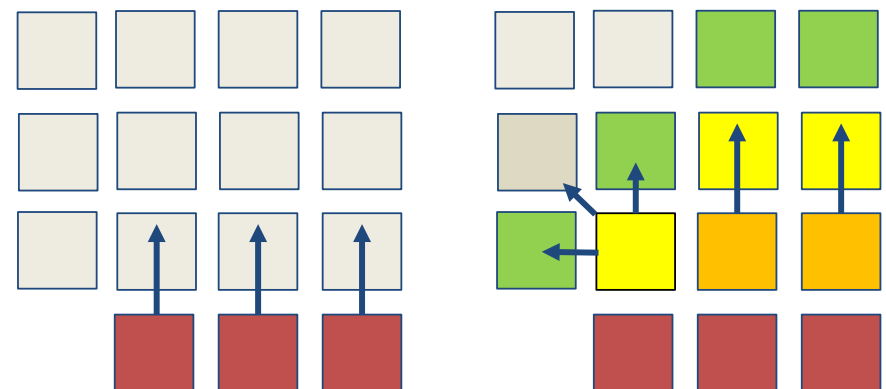
In non-discrete reaction-diffusion system, geometric constriction or dilation is the only way to cause UNIFORM wavefront to curve



Ciaccio et al., 2017



**Source = Sink**  
No curvature

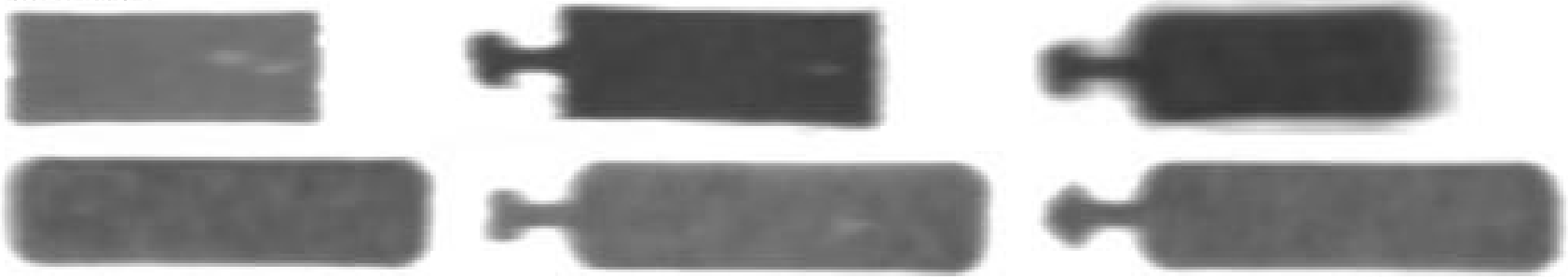


**Source < Sink**  
Concavity Forms

# Evidence of Boundary Speedup in Tissue Chips

Is uniform wavefront truly the relax state for propagating AP wavefront?

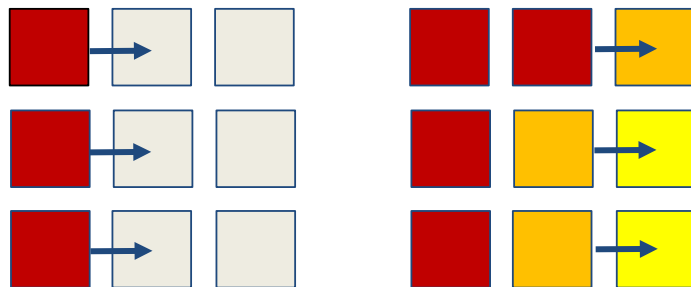
0.0000s



Lee et al., 2017

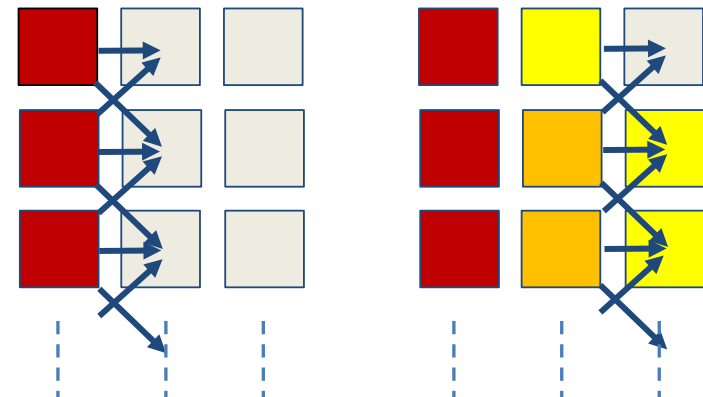
Observation: No. Evidence of lateral boundary speed up occurs despite no severe geometry pinching

Hypothesis 1



Extracellular Induced Short Circuit

Hypothesis 2

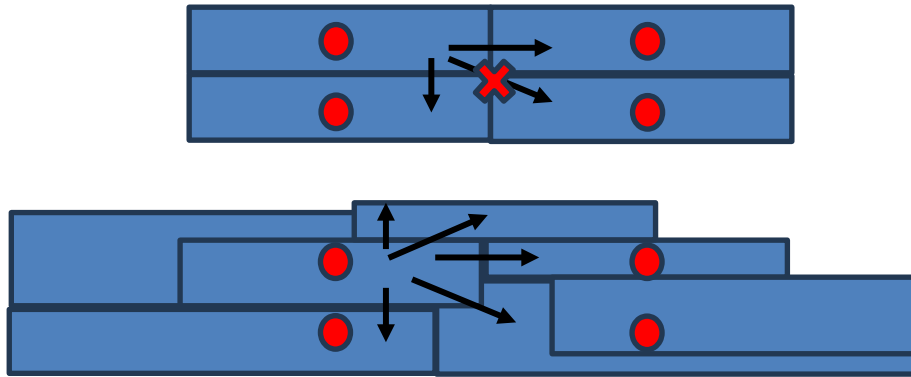


Local source-sink mismatch

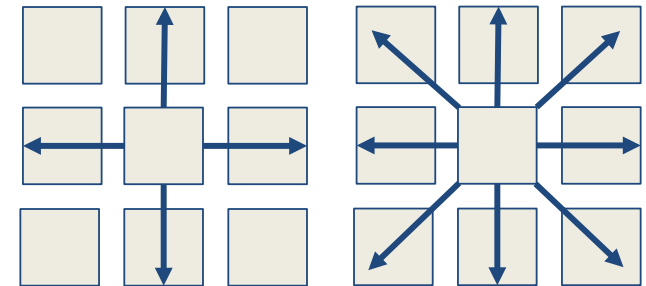
# Discrete Cardiac Connective Patterns

Numerical analysis are intrinsically discrete. What discrete features creates imbalance?

## Various Tissue Connection Networks



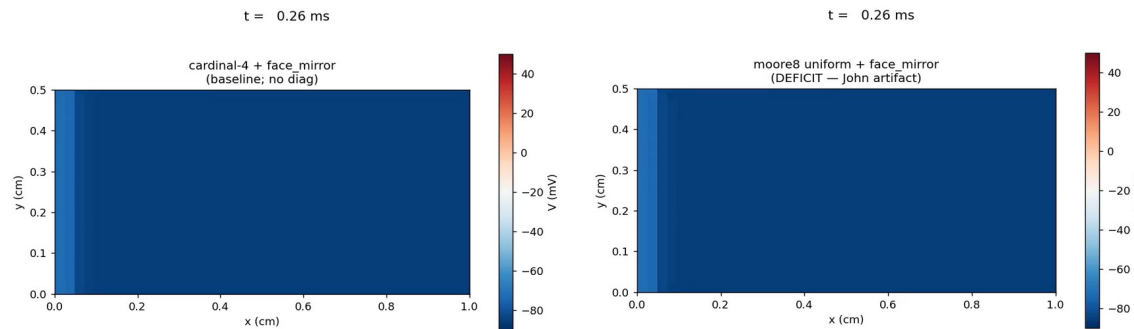
## Different Directional Connective Patterns



Cardinal

Moore

## Monodomain Simulation w Different Connective Patterns



Cardinal

Moore

Modeling cardiac tissue networks on structural grid requires diagonal connections

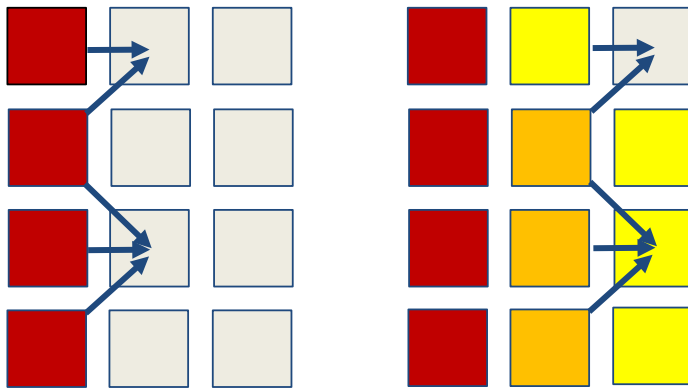
Adding diagonal connections could lead to boundary speed changes

Where is the “source” of this source-sink mismatch?

# Streaming vs Receiving

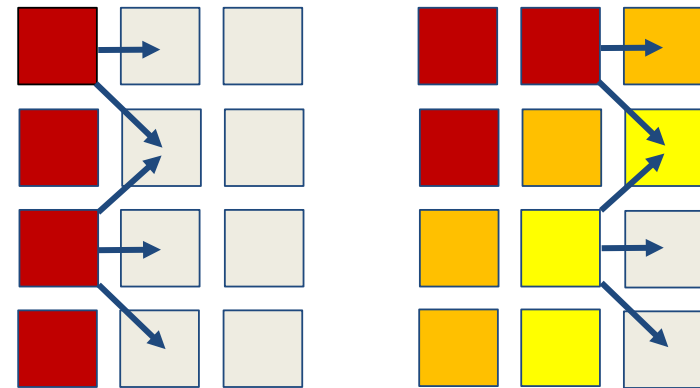
The lateral discrete source-sink mismatch occurs when there is an imbalance between receiving less vs draining less compared to the center cells.

## Effect 1: Boundary cells receive less source



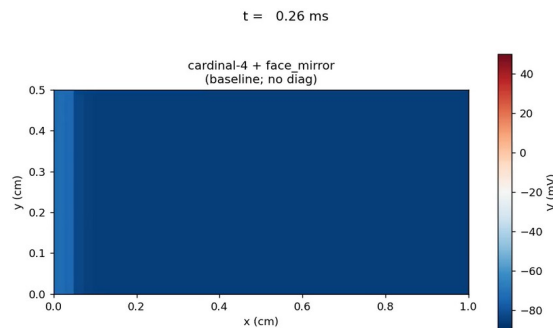
Favors Crescent Wavefront

## Effect 2: Boundary cells stream to less sink

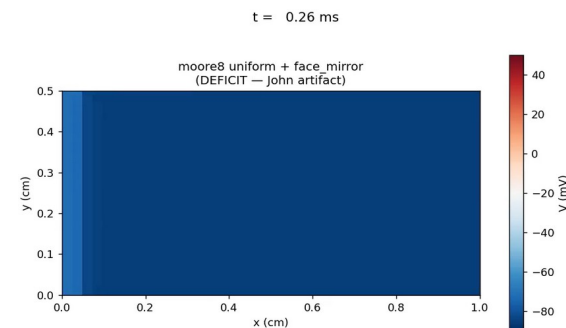


Favors Inverse Crescent Wavefront

Can we control which effect wins? Does Moore connection have a bias towards effect 1?



Cardinal



Moore

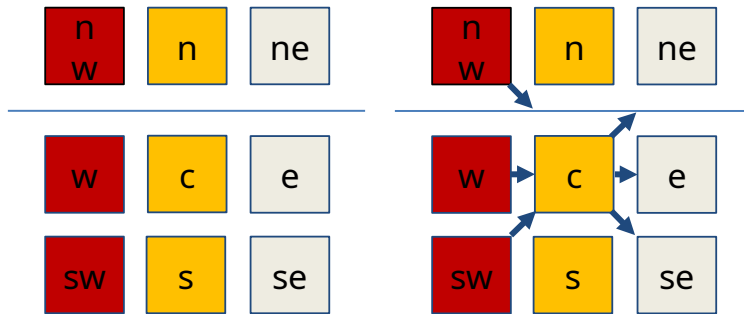




# Streaming vs Receiving Cont.

We can control which effects dominate by focusing on how boundary conditions are enforced. In cardiac simulation, we usually impose a no-flux boundary (but only to the normal direction).

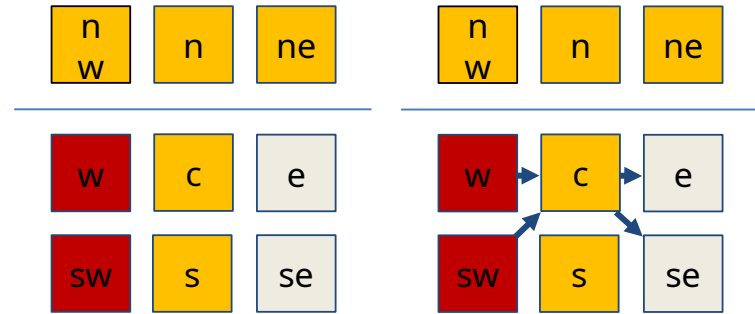
Iso padding the ghost cells



Favours Uniform Wavefront



Face mirror padding the ghost cells



Favors Crescent Wavefront

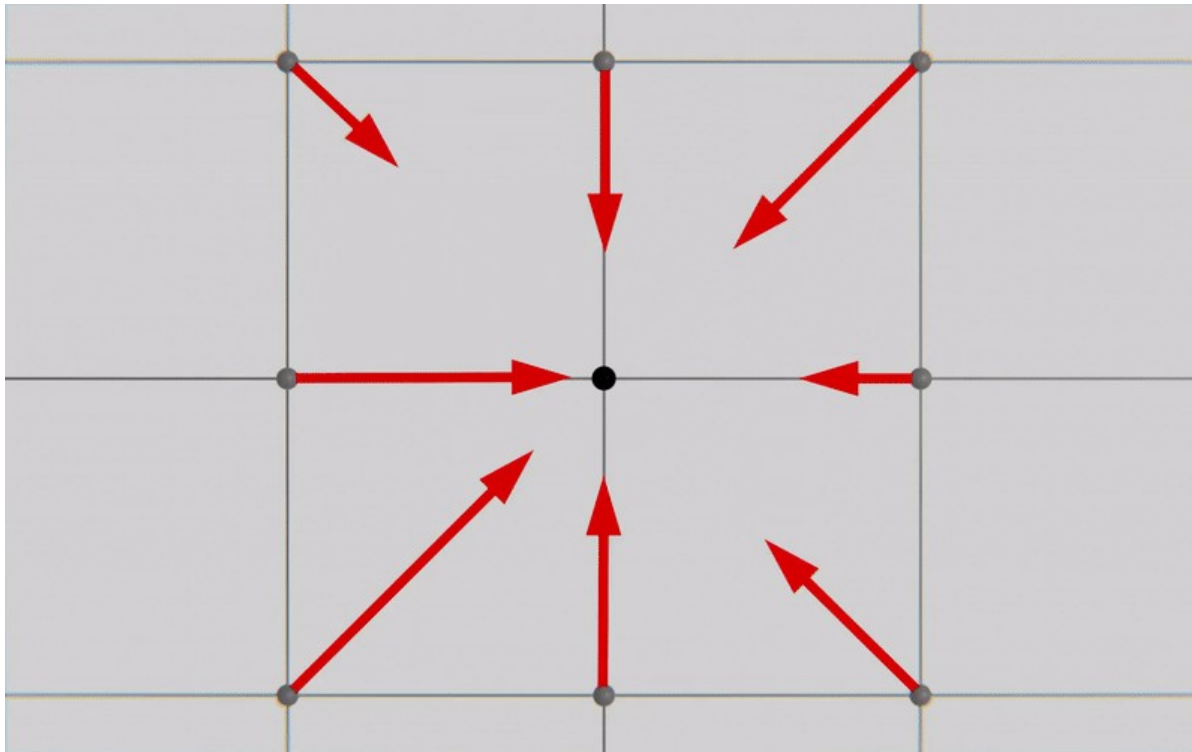


There must be a boundary condition formulation that favors steady state inverse crescent wavefront  
Issue: Monodomain simulation has limitations regarding different combo of boundary conditions



# LBM Engine Allows Control Over Directional Streaming

We need LBM: A simulation that tracks phase space instead of positional space



Medium

**Positional Encoding  
(Monodomain)**  $V(x, t)$

What is the voltage at  
location  $x$  at time  $t$

**Phase Encoding (LBM)**  
 $f(x, c, t)$

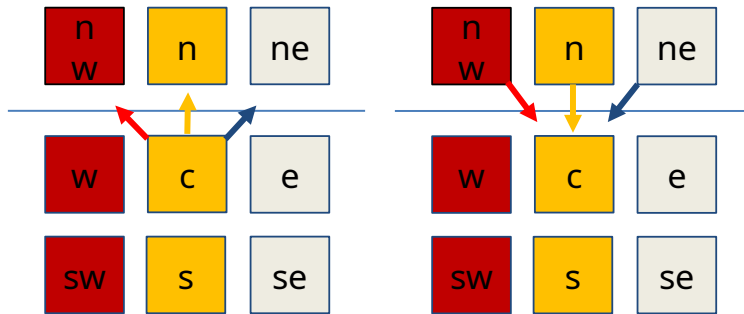
What is the voltage density at location  $x$   
with a velocity of  $c$  and at time  $t$



# Inverse Crescent Formation Supplemental

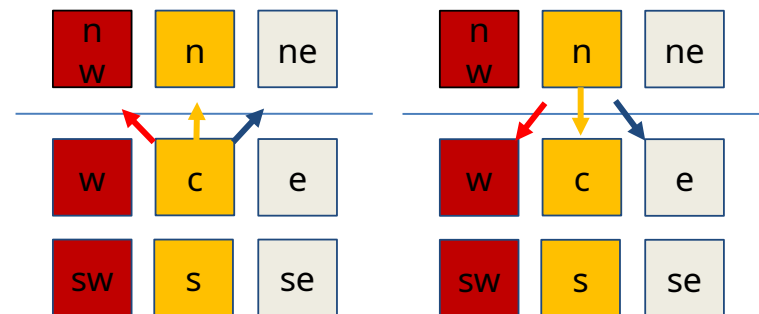
In LBM, you have direct control over the reflected direction and target of the ghost cell

Half-Way Bounce Back



Favours Crescent Wavefront

Specular Bounce Back



Favours Uniform Wavefront

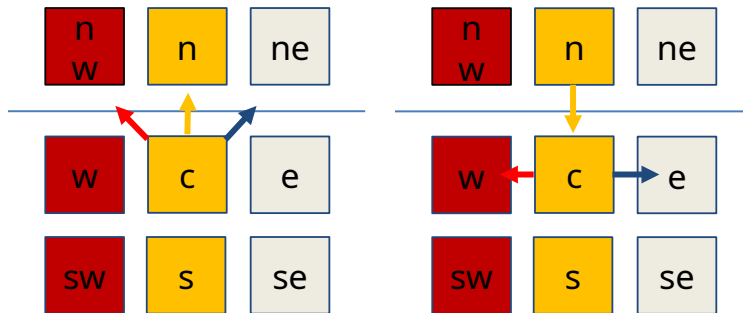




# Inverse Crescent Formation Supplemental

What if we horizontally stream back the bounce to the neighbor cells

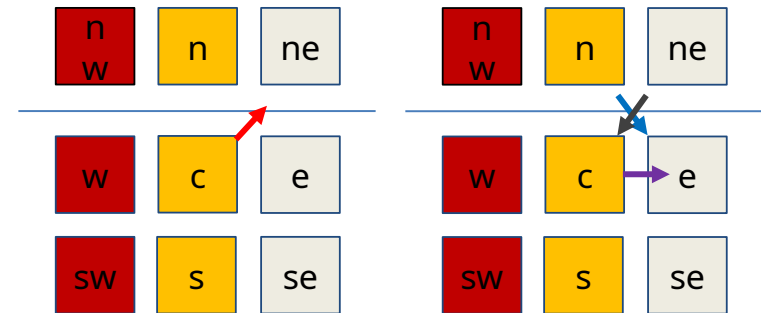
Horizontal Bounce Forward



Favours Inverse Crescent Wavefront



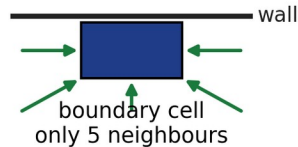
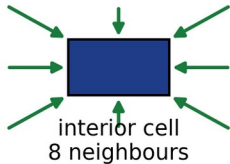
Combined Bounce Back



Weighted Combination fo HBB, Specular, and Horizontal  
Full Control Over Steady State Wavefront



# Boundary Rules Set the Crescent



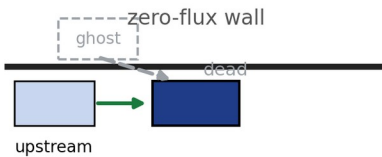
**Fewer neighbours → two competing effects:**

Effect 1 — receives less source (fewer inbound channels) → tends to SLOW DOWN

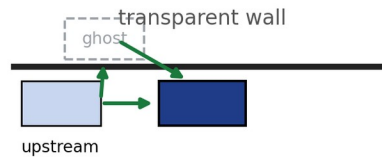
Effect 2 — streams to less sink (less downstream load) → tends to SPEED UP

*The boundary condition decides which effect wins.*

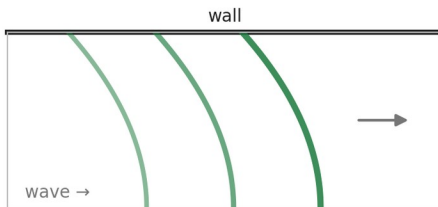
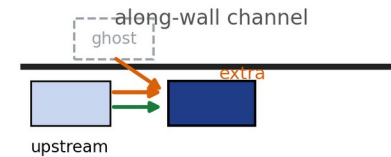
**face\_mirror ≡ HBB**



**specular ≡ face\_mirror\_iso**

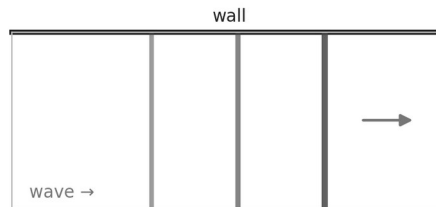


**horizontal redirect (new)**



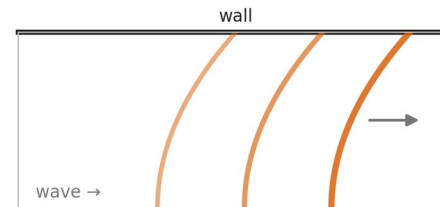
**FORWARD crescent  
boundary LAGS**

Off-grid diagonal returns the cell's OWN value → no upstream source through it.  
Effect 1 wins: boundary receives less.



**FLAT wavefront  
no bias**

Off-grid diagonal pulls a REAL upstream neighbour's value → source restored.  
Effects balance: no curvature.



**INVERSE crescent  
boundary LEADS = SPEEDUP**

Diagonal mass streamed along the wall to the next cell → wall channel races ahead.  
Effect 2 wins: boundary leads.

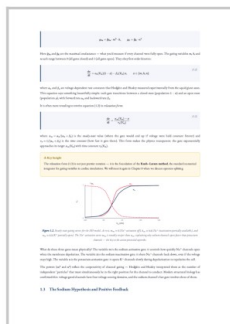
**Takeaway:** one source-sink imbalance, three wall rules — face-mirror / HBB starves the edge (forward crescent), specular is neutral, and the new horizontal-redirect over-feeds it (inverse crescent = boundary speed-up).



# A Visual Cardiac-Modeling Textbook

20 chapters · 4 parts · 139 pp · Single-Cell → Monodomain → Bidomain → LBM

*Feynman-style; equations verified against Engine V5.4 code*



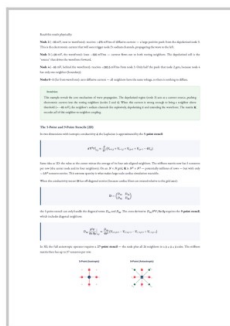
Part I — Hodgkin-Huxley  
& gating kinetics



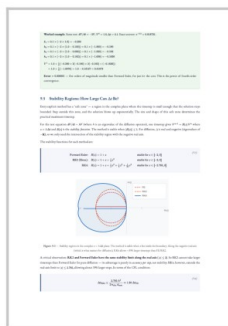
Part I — Calcium handling  
(Intuition callout box)



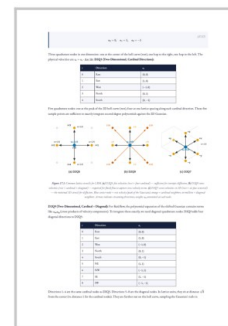
Part I — O'Hara-Rudy 2011  
ionic model



Part II — Monodomain eqn  
& spatial discretization



Part II — Diffusion solvers  
(explicit / implicit)



Part III — Bidomain:  
matrices & face stencils

*Self-authored reference, ion-channel kinetics → tissue-scale solvers — the conceptual backbone for all project engines.*

**Takeaway:** a self-authored 139-page visual textbook (ion channels → bidomain solvers, equations verified against Engine V5.4) is the conceptual backbone behind the project's engines.



# Future Goals

- **Fourier Transform Encoding of Geometric Infarct Shapes**
- **MLDO pipeline for worst geometry prediction**
- **Completion of handoff materials**
  - **Claude MCP for Monodomain, Bidomain, and LBM**
  - **Computational Cardiac-electrophysiology Textbook**